COMSATS University Islamabad
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PROJECT REPORT

Wi-Fi Controlled RC Car using ESP8266

Hardware / IoT / Applied Physics Project

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1. Abstract

This report presents a Wi-Fi Controlled RC Car built using ESP8266 NodeMCU and the L298N motor driver. The project demonstrates wireless control, motor driving, circuit wiring, battery-powered operation, and basic embedded systems concepts. The car is controlled through a mobile/web interface connected to the ESP8266 access point, allowing movement commands such as forward, backward, left, right, stop, diagonal movement, and speed control.

2. Introduction

A Wi-Fi controlled robotic car combines electrical hardware, programming, and wireless communication. The ESP8266 works as the main controller and creates a Wi-Fi access point. A phone connects to this network and sends commands to the car. The L298N motor driver receives control signals from ESP8266 and drives the motors according to selected direction and speed.

3. Objectives

- Design and assemble a functional battery-powered RC car.
- Control the car wirelessly using ESP8266 Wi-Fi access point mode.
- Use the L298N motor driver to control motor direction and speed.
- Implement movement commands for forward, backward, left, right, diagonal movement, and stop.
- Understand the link between programming, electronics, and applied physics concepts.
- Document the project using source code, images, wiring diagram, and GitHub repository files.

4. Components Used

Component	Purpose / Function
ESP8266 NodeMCU	Main controller; creates Wi-Fi access point and receives commands.
L298N Motor Driver	Controls motor direction and speed using input and enable pins.
Four DC Motors	Provide movement to the car.
18650 Battery Pack	Supplies power to the motor driver and system.
Car Chassis and Wheels	Mechanical structure for motors, board, driver, and battery.
Jumper Wires	Signal and power connections.
Mobile/Web Interface	Sends movement and speed commands to ESP8266.

5. System Design

The system is divided into the control unit, motor driving unit, power unit, and user interface. The ESP8266 receives commands through Wi-Fi and sends signals to the L298N driver. The L298N then controls the DC motors according to the command. The battery pack provides power to the motor driver and car hardware.

Module	Work Done
Control Module	ESP8266 runs the program, creates Wi-Fi network, receives commands, and controls pins.
Motor Driver Module	L298N receives ESP8266 signals and supplies required motor current.
Power Module	18650 battery pack powers the motors and driver circuit.
User Interface Module	Mobile/web interface sends command values such as F, B, L, R, S and speed numbers.

6. Mobile Control Interface

The mobile interface connects to the ESP8266 access point and sends movement commands through the local IP address 192.168.4.1.

Figure 1: Mobile controller interface for the RC car.

7. Wiring Diagram and Connections

The wiring diagram shows how ESP8266 is connected to the L298N motor driver, battery pack, and motors. A common ground connection is important so that the ESP8266 signals are correctly understood by the motor driver.

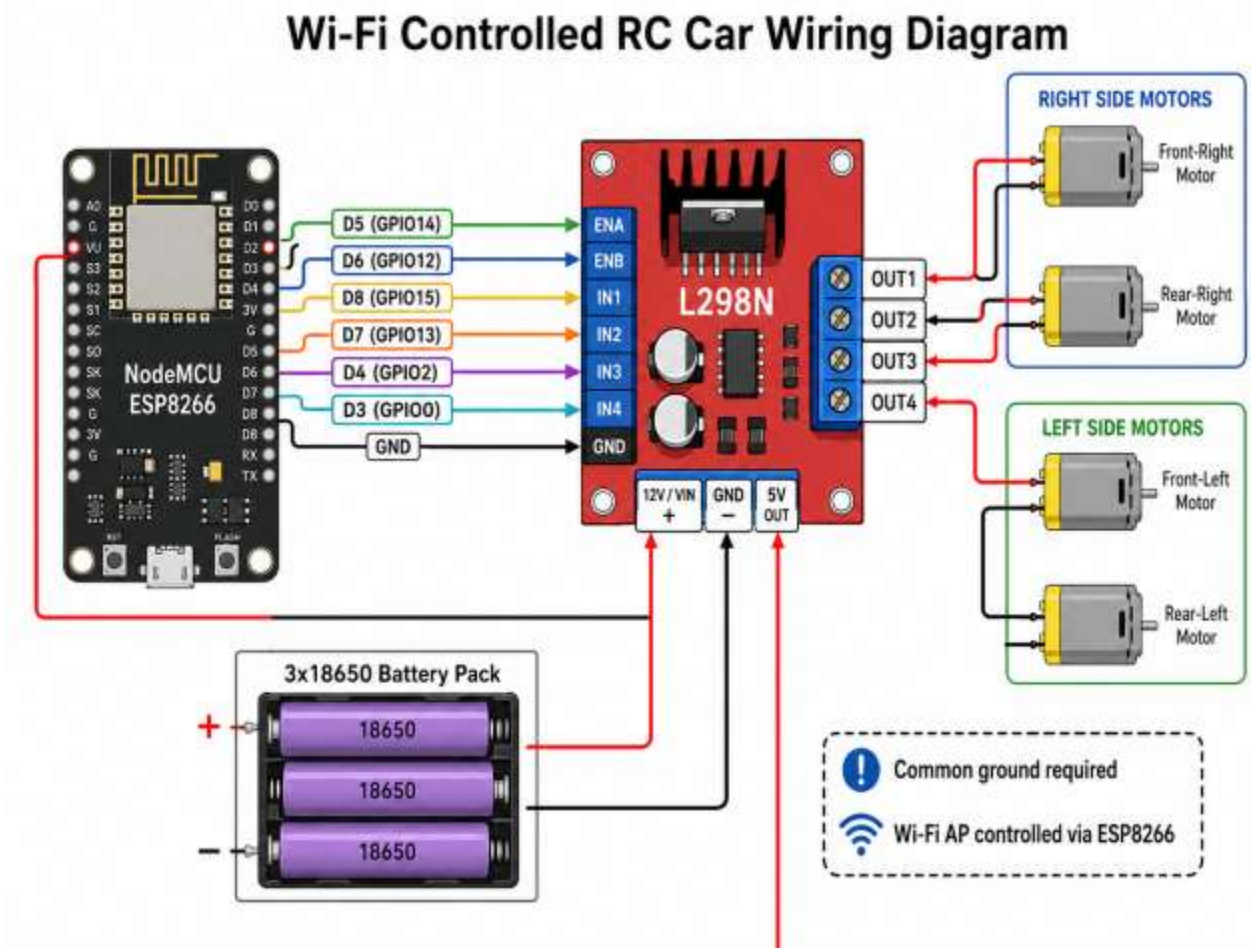

Figure 2: ESP8266 and L298N wiring diagram.

ESP8266 Pin	L298N Pin	Purpose
D5 / GPIO14	ENA	Right motor speed control
D6 / GPIO12	ENB	Left motor speed control
D8 / GPIO15	IN1	Right motor input 1
D7 / GPIO13	IN2	Right motor input 2
D4 / GPIO2	IN3	Left motor input 3
D3 / GPIO0	IN4	Left motor input 4
GND	GND	Common ground connection

8. Software Design

The program is written for NodeMCU ESP8266 using the Arduino IDE. It includes ESP8266 Wi-Fi and web server libraries. The ESP8266 creates a Wi-Fi network and starts a web server on port 80. The mobile interface sends a parameter named State, and the program checks this command to perform movement.

8.1 Main Libraries

- ESP8266WiFi.h - configures ESP8266 Wi-Fi access point.
- WiFiClient.h - supports network communication.
- ESP8266WebServer.h - creates the web server that receives control commands.

8.2 Command Mapping

Command	Action
F	Move forward
B	Move backward
L	Turn left
R	Turn right
I	Forward-right movement
G	Forward-left movement
J	Backward-right movement
H	Backward-left movement
S	Stop motors
0-9	Adjust motor speed

8.3 Important Functions

Function	Use
setup()	Initializes pins, starts Serial Monitor, configures Wi-Fi AP, and starts the server.
loop()	Handles client requests and checks the State command.
goAhead() / goBack()	Moves the car forward or backward.
goLeft() / goRight()	Changes motor directions to turn the car left or right.
stopRobot()	Stops all motors by setting motor inputs low and speed to zero.
HTTP_handleRoot()	Sends a simple OK response to the client.

9. Working Flow

1. The battery pack supplies power to the motor driver and control circuit.
2. ESP8266 starts in Wi-Fi Access Point mode using the network name defined in the code.
3. The mobile phone connects to the ESP8266 Wi-Fi network.
4. The mobile/web interface sends movement commands to the ESP8266 through the web server.
5. ESP8266 reads the State command and calls the related movement function.
6. L298N receives pin signals from ESP8266 and drives the motors.
7. The car moves forward, backward, left, right, diagonally, or stops based on the command.
8. Speed values from 0 to 9 change the PWM value applied to the enable pins.

10. Testing and Results

Test	Result
Wi-Fi connection	ESP8266 access point was visible and mobile device connected successfully.
Forward/backward movement	The car responded to forward and backward commands.
Left/right movement	The car turned by changing motor directions.
Stop command	Motors stopped when stop command was sent.
Speed control	Different speed values changed motor speed using PWM.

11. Applied Physics and Engineering Concepts

- Electrical energy from the battery is converted into mechanical motion through DC motors.
- Motor speed changes because of PWM control through enable pins.
- The L298N driver supplies motor current because ESP8266 pins cannot drive motors directly.
- Friction between tyres and surface helps the car move and turn.
- Battery voltage, current capacity, and motor load affect performance and running time.
- Correct polarity and common grounding are important for stable operation.

12. Challenges Faced

- Connecting ESP8266 pins correctly with L298N motor driver inputs.
- Managing motor power separately from control signals.
- Ensuring all ground connections were common.
- Understanding motor direction changes using IN1, IN2, IN3, and IN4 pins.
- Adjusting motor speed values for smooth movement.
- Organizing wires and components on a compact chassis.

13. Future Improvements

- Add a rechargeable battery charging module for safer charging.
- Add obstacle detection using ultrasonic or IR sensors.
- Improve the mobile interface with a cleaner dashboard.
- Add a camera module for live video control.
- Add speed feedback and battery level monitoring.
- Design a neater PCB-based wiring layout.

14. Conclusion

The Wi-Fi Controlled RC Car using ESP8266 is a successful practical project that combines programming, electronics, wireless communication, and applied physics. The project demonstrates how a microcontroller can receive wireless commands and control motor movement through a driver circuit. It helped the team understand ESP8266 programming, L298N motor control, battery-powered circuits, PWM speed control, wiring, and troubleshooting. Overall, the project is a useful foundation for robotics, IoT, and embedded systems learning.

15. GitHub Repository Contents

- README.md - project overview and documentation.
- WiFi_Controlled_RC_Car_ESP8266.ino - source code uploaded to ESP8266.
- RC car images - hardware photographs and generated model images.
- Mobile controller screenshot - image of the control interface.
- Circuit diagram - wiring diagram for ESP8266, L298N, motors, and battery pack.
- Project poster - visual project thumbnail for LinkedIn/GitHub.

Appendix A: Source Code File

The complete source code is included in the GitHub repository as WiFi_Controlled_RC_Car_ESP8266.ino.